" if you want to manage it Measure it "

IOTAFLOW
SYSTEMS PVT. LTD

- Landfill site
- Waste water treatment
- Bio-methane plant
- CDM project
- Anaerobic digestion
- Other fermentation processes

UF- BioSonic series

ULTRASONIC BIOGAS FLOWMETER

www.iotaflow.com

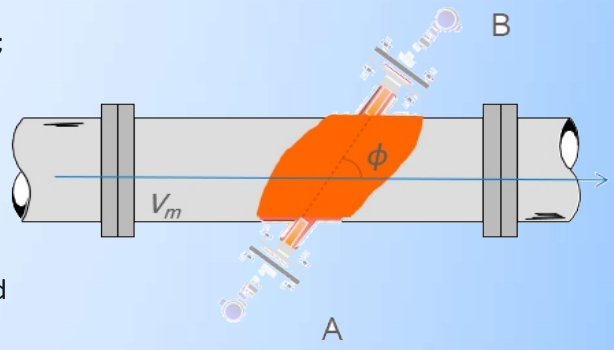
The UF- BioSonic series flow meter using latest ultrasonic transit-time technology to measure flow. With advantages of no moving parts, no pressure loss, wide rangeability/turn down ratio, corrosion resistance, the meters are capable of providing reliable long-term accuracy and require no routine maintenance. It is also designed for wet and corrosive working condition, with biogas composition reading at real time. Compared with vortex, swin and rotany flow meter, the BF-3000 flow meter is able to supply the optimized solutions for biogas, without pressure drop, wear and corrosive problems.

Ultrasonic flow meter works under the principle of the relations between sonic wave travel time in flow media and flow rate. The actual speed of sonic wave in fluid, usually speaking, is combined of flow speed of sonic wave (Cf) in static condition of media and component of average axial flow velocity of liquid(Vm) in travel direction of sonic wave. As below picture shows, upstream and downstream travel time is calculated as formula 1 and 2.

Formula 1 $$t_{down} = t_{AB} = \frac{L}{(c_f + v_m \cos\phi)}$$

Formula 2 $$t_{up} = t_{BA} = \frac{L}{(c_f - v_m \cos\phi)}$$

t_{up} — upstream travel time of ultrasonic wave in fluid;
t_{down} — downstream travel time of ultrasonic wave in fluid;
L — length of sound channel;
Cf — travel velocity of ultrasonic wave in fluid;
Vm — average axial flow velocity of fluid ;
ϕ — angle of sound channel;



According to the formula 1 and formula 2, the flow speed of fluid is calculated as follow.

$$v_m = \frac{L}{2\cos\phi}\left(\frac{1}{t_{down}} - \frac{1}{t_{up}}\right)$$

Volume flow rate is the flow velocity of fluid times orifice area A. Formula is as follow

$$q_v = A v_m$$

SPECIFICATIONS

Technical parameters									
Applicable pipeline	DN32	DN50	DN80	DN100	DN125	DN150	DN200	DN250	DN300
Flow range (m³/h)	1.3-65	3.2-160	4.8-240	8-400	12-600	16-800	32-1600	48-2400	64-3200
Interface	1"1/4	2"	Convex flange						
Accuracy range	≤±1.5%								
Working pressure	0-200kPa								
Working temperature	-10°C-40°C								
Humidity	≤99%								
CH_4 composition	Range: 30-100%Vol, Accuracy: ±5%FS								
Power supply	External power supply : Safety Barrier Ui=5V Li=100mA (Main power supply) Internal battery:3.6V LS26500 (Backup power supply)								
Communication	RS485								
Protection class	IP65								

FEATURES

- Ultrasonic technology with no moving parts,less pressure drop.
- Methane content measurement function.
- Temperature & pressure compensation Optional.
- Easy to install, routine maintenance free.
- Strong corrosion resistance from high H2S and moisture.
- Long lifetime, stable performance.
- Built-in industrial lithium battery.
- Wide High/Low-flow ratio for precision measurement.

Contact Us

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